

Cloud Security Essentials TASK -4

- 1. Encrypt a sample text file (such as your Spotify playlist export) using any open-source tool like VeraCrypt or 7-Zip, then decrypt it and verify the contents match the original.

Hint: Use password protection and AES-256 encryption if available.**

Ans :

- Install 7-Zip on your computer.
- Create a sample text file named playlist.txt.
- Add sample content, for example
 - Spotify Playlist
 - Blinding Lights
 - Shape of You
 - Believer
- Save the file.
- Right-click playlist.txt.
- Select 7-Zip → Add to archive...
- Choose the archive format as 7z (or ZIP).
- In the Encryption section:
 - Enter a strong password.
 - Re-enter the password.
 - Set Encryption method to AES-256.
- Click OK to create the encrypted archive (e.g., playlist.7z).
- Verify that the encrypted archive has been created.

- Delete or move the original playlist.txt file (optional, for testing).
- Right-click the encrypted archive (playlist.7z).
- Select 7-Zip → Extract Here (or Extract to "playlist").
- Enter the password when prompted.
- Click OK to decrypt the file.
- Open the extracted playlist.txt.
- Verify that the contents exactly match the original text.

**2. Capture network traffic while logging into your Instagram or Zomato account on a web browser using Wireshark or Fiddler, and identify whether your login data is transmitted over HTTPS.

Hint: Look for the padlock icon in your browser and check the protocol in captured packets.**

Ans :

- Install Wireshark (or Fiddler) on your computer.
- Launch the application.
- Select your active network interface (Wi-Fi or Ethernet).
- Click Start Capturing Packets.
- Open a web browser.
- Go to <https://www.instagram.com> or <https://www.zomato.com>.
- Check that the website displays the padlock icon in the address bar, indicating a secure HTTPS connection.
- Log in to your account using your username and password.

- After logging in, return to Wireshark and stop the packet capture.
 - In Wireshark, apply the display filter
 - Tls
 - Inspect the captured packets and confirm that the traffic uses TLS over port 443, which indicates HTTPS.
 - Verify that your username and password are not visible in plain text, because the data is encrypted.
- 3. Use the open-source tool 'Nikto' or 'OWASP ZAP' to run a basic vulnerability scan on a local web app (such as a demo site running on localhost) and record at least 3 security issues found.**

Ans :

- Install OWASP ZAP on your computer.
- Start your local demo web application (e.g., <http://localhost:8080>).
- Open OWASP ZAP.
- Select Automated Scan from the home screen.
- Enter the target URL
- <http://localhost:8080>
- Click Attack (or Start Scan).
- Wait for the scan to complete.
- Open the Alerts tab to review the detected security issues.
- Record at least three vulnerabilities found.

Issue	Risk Level	Description
Missing Security Headers	Medium	HTTP security headers such as X-Frame-Options or Content-Security-Policy are missing, making the application more vulnerable to attacks.
Cookie Without Secure Flag	Medium	Cookies are not marked as Secure, allowing them to be transmitted over non-HTTPS connections.
Information Disclosure	Low	Server response headers reveal software or server version information that could help attackers identify potential vulnerabilities.

4. Perform a simulated security audit by reviewing the permissions of files in your cloud storage (Google Drive or OneDrive). List any files that are shared publicly or with 'Anyone with the link' and suggest how to restrict access for better security.

Ans :

- Sign in to your Google Drive or OneDrive account.
- Open the My Drive (Google Drive) or Files (OneDrive) section.
- Review the sharing settings for your files and folders.
- Identify any files marked as:
 - Anyone with the link
 - Public

- Anyone on the internet
- Open the Share settings for each publicly shared file.
- Record the file name and its current sharing permission.
- Change the sharing setting from Anyone with the link or Public to one of the following:
 - Restricted (Only specific people) (Google Drive)
 - Specific people (OneDrive)
- Save the updated sharing permissions.
- Repeat the process for all publicly accessible files.

File Name	Current Permission	Security Risk	Recommended Action
ProjectReport.pdf	Anyone with the link	Anyone with the link can access the file.	Change access to Restricted or Specific people.
Resume.docx	Public	File is accessible to anyone on the internet.	Remove public access and share only with intended users.
Photos.zip	Anyone with the link	Unauthorized users may access if the link is shared.	Restrict access to authorized users only.

5. After running a vulnerability scan, pick one reported issue (such as 'Directory Listing Enabled') and write a short action plan describing how you would fix this issue in a real cloud environment.

Ans :

- Issue :
- The vulnerability scan reported "Directory Listing Enabled." This means users can view the contents of a web server directory if no default page (such as index.html) exists. Attackers may use this information to discover sensitive files or application structure.
- Action Plan :
- Disable directory listing in the web server configuration (Apache, Nginx, or IIS).
- Add a default page (such as index.html) to directories that should not display file listings.
- Restrict access to sensitive folders using proper permissions and access control rules.
- Remove unnecessary files (backup files, logs, and temporary files) from public directories.
- Apply the principle of least privilege so only authorized users can access sensitive directories.
- Run another vulnerability scan after making the changes to confirm that the issue has been resolved.
- Monitor server logs regularly to detect any unauthorized attempts to access restricted directories.

